Mining Chemical Signatures and Modeling Quality Outcomes: An Integrated Approach to Binary Wine Classification

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Abstract

Wine quality assessment plays a critical role in the guide of production decisions, pricing strategies and market positioning. However, traditional sensory evaluation by human tasters is subjective and often inconsistent, which has increased the demand for objective and data driven methods. This study applies machine learning techniques to classify wines from a combined red and white Vinho Verde data set using 11 physicochemical properties. To allow for a clearer interpretation, the continuous quality scores were grouped into two classes that is Good (7) and Bad (6). After merging and cleaning the datasets, preprocessing steps included removing multicollinearity, standardizing numerical variables, and performing an 80/20 train test split. Three classification models Binary Logistic Regression, Decision Tree and Random Forest were implemented to evaluate the effectiveness of linear and nonlinear approaches and to prove which model performed well. Logistic Regression served as a transparent baseline model and identified alcohol, sulphates and chlorides as strong predictors, though its performance favored the majority class. The Decision Tree improved well on the interpretation and captured nonlinear relationships with alcohol and chlorides emerging as the most important features but struggled to classify the minority class which was good. The Random Forest showed the best overall performance accuracy which is 0.82 benefiting from bagging and random feature selection to reduce variance and stabilize predictions. The importance of features highlighted alcohol, density and chlorides as the primary drivers of wine quality in the ensemble. Overall, the findings d that tree based models provide stronger predictive capability than a linear baseline, particularly in datasets with complex relationships. However, class imbalance remains a major challenge. These results emphasize the value of integrating machine learning into wine quality evaluation while showing the need for further techniques such as resampling or cost sensitive learning to better capture minority high quality wines.

1 Introduction

Wine quality evaluation plays a major role in the wine industry, shaping production choices, pricing and how wines are positioned in the market. Traditionally, wine quality is judged by expert tasters who rely on sensory characteristics such as aroma, flavor balance, acidity and overall texture. While this approach is valuable, it is also subjective that different tastes may give different ratings depending on their experience, conditions or preferences. Because of these inconsistencies there is growing interest in using data driven methods that rely on clear measurable chemical properties to support or complement human judgment and as supported by[1] who says that wine classification is a difficult because taste is the least understood of the human senses. Recent advances in data mining and machine learning have made it possible to analyze the chemical variables of wine and uncover patterns associated with quality scores as depicted by [2] who says advances in information technologies have made it possible to collect , store and process massive often highly complex datasets as well of improving decision making and optimize chances of success. Physicochemical variables such as fixed acidity, volatile acidity, residual sugar, chlorides, sulphates, pH, density, alcohol content and sulfur are routinely measured in wine making and provide objective indicators of wine characteristics. Using these variables, machine learning models can predict wine quality more consistently while also revealing which chemical factors matter most.

In this study, we work with a combined dataset of red and white Vinho Verde wines from the UCI Machine Learning Repository. The dataset contains over 6,000 samples and includes 11 physicochemical variables along

with a continuous quality score. To support classification modeling and simplify interpretation,we grouped the quality ratings two categories good (Quality ≥ 7) or bad (Quality ≤ 6) . Before modeling, we performed several preprocessing steps merging the datasets, checking for missing values, standardizing numerical variables, and examining correlations among predictors. These steps ensured that the data was clean, consistent and ready for analysis. The goal of this research is to classify wines as either good or bad using their physicochemical properties and to determine which features play the biggest role in this classification. To accomplish this, we applied two tree based machine learning models: Decision Tree and Random Forest. Decision Trees offer clear interpretability by showing how the model splits the data, while Random Forest uses many trees together to improve accuracy and reduce overfitting. We evaluated these models using common classification metrics, including accuracy, precision, recall and confusion matrices. We also examined feature importance scores to understand which chemical factors influenced wine quality the most. Overall, this study aims to support a more objective, data driven approach to wine quality assessment. The findings show the value of tree based machine learning models in predicting wine quality and highlight the physicochemical variables that consistently contribute to better or worse quality outcomes.

1.1 Background

Wine quality plays a central role in the wine industry because it influences everything from pricing and marketing to consumer preference and production decisions. Winemakers depend on quality assessments to monitor fermentation, adjust chemical balances and maintain consistency across batches. However, traditional wine evaluation is carried out by human tasters whose judgments can be subjective. Factors such as personal preference, experience and even testing conditions can cause the same wine to receive different scores at different times. This variation makes it difficult for producers to rely solely on sensory evaluation when making important decisions. Because of this subjectivity, data driven methods have become increasingly important in supporting or refining human assessments. Physicochemical measurements such as acidity, alcohol content, pH, density and sulfur dioxide levels provide objective indicators of wine characteristics that do not change based on who is evaluating the wine. By applying statistical and machine-learning models to these measurable features we can estimate wine quality more consistently and identify which chemical properties contribute most to perceived quality. This modelling approach offers a reliable, repeatable way to complement human judgment and helps producers make informed decisions grounded in both science and sensory experience

1.2 Objective

The objective of this study is to predict wine quality using measurable physicochemical properties and evaluate how effectively machine learning models can capture the underlying patterns in these features. This is important because accurate and objective quality prediction supports winemakers in making informed production decisions, improves consistency and reduces reliance on subjective sensory evaluations. The project also aims to identify the variables that have the strongest influence on wine quality and compare the performance of different modeling approaches, including binary regression, decision trees and random forest methods. Through these goals, the study seeks to provide a more objective and reliable way of assessing wine quality beyond sensory evaluation.

1.3 Variables Description

The dataset contains 11 physicochemical measurements that describe the chemical makeup of each wine sample. These features capture important aspects of acidity, sweetness, composition and stability, all of which influence how wine tastes and how it is perceived by tasters. Fixed acidity refers to the non-volatile acids that remain in the wine after fermentation and contribute to its overall structure. Volatile acidity captures the level of acetic acid if this becomes too high, it can create a sharp, vinegar like smell. Citric acid is naturally present in small amounts and can add a fresh bright flavor to the wine. Residual sugar measures how much sugar remains after fermentation, which affects sweetness. Chlorides represent the salt content, while free sulfur dioxide and total sulfur dioxide indicate how well the wine is protected against oxidation and microbial growth. Density reflects the balance between sugar and alcohol in wine and pH provides a general measure of acidity with lower values indicating a more acidic wine. Sulphates are additives that help with preservation and contribute to the wine's stability. Alcohol shows the percentage of ethanol in wine and is often one of the strongest indicators of overall quality. Finally, the quality variable is the target label.

2 Methods

2.1 Dataset Source

The dataset used in this project was obtained from OpenML a well-known platform that shares curated datasets for machine learning research. OpenML provides clean well-structured datasets that are easy to access and work with, which makes it a reliable source for academic studies. The wine quality dataset we selected is publicly available and frequently used in research which shows that the data is credible and suitable for the type of analysis carried out in this study.

2.2 Data Collection and Pre-Processing

The dataset was made by the combination of red and white wines which are variants of the Portuguese "Vinho verde wine".The data was collected from May 2004 to February 2007 and it only considered original samples from CVRVV[3]. The main goal of CVRVV is to ensure the consistent quality and authenticity of Vinho verde wine. CVRVV utilizes a computerized system (ilab) to manage the collection of data for different wine samples.The collection includes physiochemical tests from the laboratory and panel sensory analysis.The dataset contains 1597 instances of red wine and 4900 instances of white wine, It also includes 11 physiochemical properties and sensory scoring, a median of 3 evaluations made by wine experts [3]. Due to privacy and security concerns, no information was available regarding branding, grape types and selling price. The explanatory variables in the dataset include: fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH 10, sulphates, alcohol and quality as the response variable.

The response variable is a score from 0 (very bad) to 10 (excellent), conducted under a minimum of 3 wine experts[3].Preprocessing steps was performed to ensure consistent formats of all chemical variables. Missing values were checked and no null variables were detected. The numerical variables were standardized to reduce the influence of differing measurement ranges among the physicochemical attributes, which is suitable for magnitude sensitive models. For the classification task at hand the quality scores were categorized into good wine (Quality ≥ 7) and bad (Quality ≤ 6) with the sole goal of comparing the performance of classifiers [4]. The distribution of both classes is show by bar graph 2a, there is a potential high number of low quality wine an evidence of class imbalance. For machine learning analysis the data was split into training data 80% and testing data 20%. This study used smote for handling class imbalance, specifically on the training data. Table 1 shows descriptive statistics, including mean, minimum, maximum, and standard deviation for each variable to examine their central tendency and dispersion

Table 1: Descriptive Statistics for Physicochemical Properties of Wine Samples

Variable	**Mean**	**Std**	**Min**	**25%**	**Median**	**75%**	**Max**
fixed.acidity	7.215	1.296	3.8	6.4	7.0	7.7	15.9
volatile.acidity	0.340	0.165	0.08	0.23	0.29	0.40	1.58
citric.acid	0.319	0.145	0.00	0.25	0.31	0.39	1.66
residual.sugar	5.443	4.758	0.6	1.8	3.0	8.1	65.8
chlorides	0.056	0.035	0.009	0.038	0.047	0.065	0.611
free.sulfur.dioxide	30.525	17.749	1	17	29	41	289
total.sulfur.dioxide	115.745	56.522	6	77	118	156	440
density	0.995	0.003	0.987	0.992	0.995	0.997	1.039
pH	3.219	0.161	2.72	3.11	3.21	3.32	4.01
sulphates	0.531	0.149	0.22	0.43	0.51	0.60	2.00
alcohol	10.492	1.193	8.0	9.5	10.3	11.3	14.9

2.3 Feature Selection and Correlation Matrix

Feature selection is important in machine learning and data mining fields as it contributes to model efficiency and prediction capability. Utilizing all input features for predicting your response variable is not ideal as it increases computational complexity and reduces model accuracy. Predictors that are highly correlated might lead to high variance or distorting the coefficient estimates.Selecting the relevant variables not only aids in model accuracy but reduction in data collection costs. Different methods of understanding the features selection have been proposed in various studies that cover this domain. The approach of feature selection is based on the particular model and assumptions that need to hold . In this study different techniques were utilized for feature selection due to the fact that the used models embodied different assumptions. The binary logistic regression is highly affected by multicollinearity as impacts the reliability of the coefficients and ultimately its output. Due to this fact, the pairwise Pearson correlation matrix was employed to investigate which a pair of variables are highly correlated with each other. The threshold value that was considered in this research was 0.6, so as to ensure consistency and clarity in analysis .The absolute correlation values greater than 0.5 were considered to be a highly correlated with each other and the decision involved dropped either of the variables. The variables that were dropped include density and free sulphur dioxide. Density was highly correlated with alcohol with an absolute value of 0.67 and free sulfur dioxide highly correlated with total sulfur dioxide with an absolute value of 0.72.

For decision trees and random forest, the correlation matrix was not employed in selection of the features. The decision tree based models handle multicollinearity as the data is split hierarchical and have a built in functinality of selecting informative predictors regardless of strong correlation between the predictors. The algorithm will select the best informative predictor that provides the best split and disregard the later. Furthermore, random forest provides the output based on the average of trees that are not affected by multicollinearity.These decision trees based models utilize all variables and split the data based on two factors. The factors are based on feature importance and impurtity reduction.

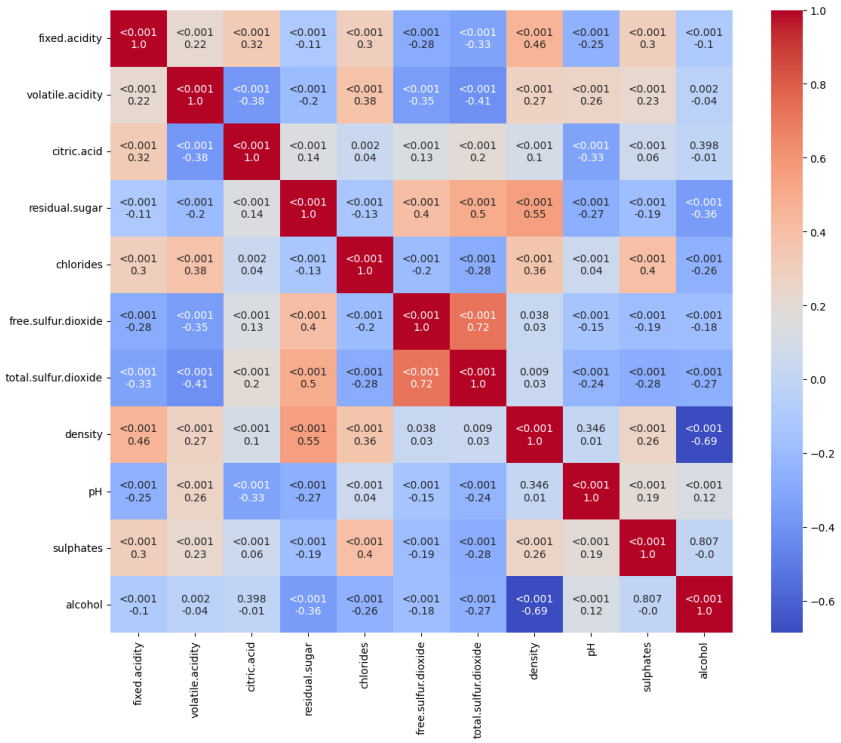


Figure 1: Correlation Matrix of the physiochemical properties of wine

2.4 Model Selection

This research utilizes a classification approach to predict the whether the particular wine is good wine (Quality ≥ 7) or bad (Quality ≤ 6) . This is a similar approach that simulates the recent study [5]. The decision was made to use binary logistic regression, decision tree and random forest models in this study.

2.4.1 Binary Logistic Regression:

In this study, binary logistic regression was used as a simple linear model to classify wine samples as good or bad based on their physicochemical characteristics.[6] described logistic regression as part of the broader family of generalized linear models, where the logit link function and maximum likelihood estimation allow for stable and reliable classification. In this study, Logistic Regression served as a simple, transparent baseline model for predicting wine quality. The central idea behind logistic regression is that it tries to draw a straight-line boundary between the two classes by learning how each input variable influences the probability of a sample belonging to the "good" class. Because the model is linear, it performs best when the two groups can be separated by a clear, consistent trend in the data. The model estimates the probability that a given wine sample belongs to the positive class using the logistic sigmoid function, which transforms a linear combination of the predictors into a value between 0 and 1. Mathematically, the model is expressed as follows

$$P(Y = 1 \mid X) = \sigma(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k)$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

A positive coefficient increases the probability of predicting the "good" class, while a negative coefficient decreases it. The model learns these coefficients by finding the values that minimize the classification error using an optimization routine such as gradient descent. Once the model computes the probability for each sample, a decision threshold often set at 0.5 is applied to convert probabilities into final class predictions. This makes logistic regression especially interpretable, as each coefficient directly reflects how a one unit change in a feature affects the log-odds of the wine being classified as good. Because logistic regression relies on a linear decision boundary, it may struggle when the relationship between features and class labels is highly nonlinear or when the two classes overlap significantly. However, despite this limitation, it remains a valuable baseline method because it is transparent, easy to interpret, and provides insight into which features have the strongest directional influence on wine quality classification. Overall, Binary Logistic Regression served as a simple yet informative model in this project. It allowed us to examine how individual physicochemical properties affect the likelihood of a wine being categorized as good or bad, offering a clear point of comparison to more flexible models like Decision Trees and Random Forest.

2.4.2 Decision Forest

Decision Trees were also applied in this study to model wine quality using physicochemical measurements. A decision tree makes predictions by repeatedly splitting the data into smaller and more homogeneous groups. At each step, the tree selects the feature and threshold that best improves the purity of the resulting groups. Purity is evaluated using entropy, which measures how mixed the classes are in each node defined as :

$$H(S) = -\sum_{i=1}^{c} p_i \log_2(p_i)$$

The feature that produces the highest Information Gain is selected for the split, and this recursive process continues until the model reaches a stopping condition.The model selects the split that provides the highest Information Gain:

$$IG(S,A) = H(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} H(S_v)$$

These splitting rules create a set of simple, interpretable if-then decisions that map physicochemical properties to a predicted wine category. Because decision trees can grow deep and fit noise in the data, they are prone to overfitting. To promote generalization, we limited the tree's depth and required a minimum number of samples before a split could occur. These regularization steps help the tree capture meaningful relationships without

memorizing the training data. Although a single tree offers clear interpretability, it can be sensitive to small changes in the dataset, which is why ensemble methods such as Random Forest often perform more consistently. Overall, the Decision Tree provided a transparent, easy to understand baseline model. It showed how the algorithm uses recursive splitting and entropy-based criteria to learn structure from the physicochemical measurements, aligning with the classical CART framework

These numbers simply reflect how clear the splits were for bad wine when the model was being trained. For good wine, the patterns were not as clear. [7] decision trees work by recursively partitioning the data into purer groups using measures such as the Gini index and can easily overfit without pruning.The tree had to go through deeper splits and the data for this class overlapped more.Therefore, the success of decision tree techniques mainly depends on several factors contributing to their performance, interpretability, and applicability to a wide range of problems. These factors include data quality, tree depth, splitting criteria, and tree pruning method. the effectiveness of decision trees is highly dependent on the training data quality. Hence, it is necessary to use clean or preprocessed data not containing missing values and outliers, which can significantly enhance the performance of the resulting models.

2.4.3 Random Forest:

Random Forest was used in this study to predict wine quality based on physicochemical measurements. Instead of relying on a single decision tree, which can easily overfit or produce unstable predictions the Random Forest algorithm builds many trees and then aggregates their outputs. By combining the predictions of multiple trees, the model becomes more stable and less sensitive to noise in the data. This structure directly addresses the high variance problem of individual decision trees.The algorithm works through a process known as bagging, where each tree is trained on a bootstrap sample meaning the dataset is randomly sampled with replacement so that every tree sees a slightly different version of the training data. In addition, at each split inside a tree the algorithm does not consider all available features. Instead, it randomly selects a subset of predictors before choosing the split that best reduces impurity. These two sources of randomness bootstrapping and random feature selection ensure that individual trees are decorrelated, which improves generalization of the overall model. Bootstrap sampling means sampling the data with replacement, the algorithm generates multiple training sets that are slightly different from each other. This type of sampling ensures reduced variance and prevents overfitting and Feature Randomization means that randomly selecting a subset of features for each tree ensures the algorithm decorrelates the trees and reduces the chance of selecting the same "best" feature for every tree. This improves the diversity and accuracy of the trees. Mathematically, the Random Forest prediction for a regression problem is computed by averaging the predictions from all trees in the ensemble, this can be expressed as :

$$\hat{y} = \frac{1}{T} \sum_{t=1}^{T} f_t(x)$$

As [8] highlighted, this averaging makes Random Forest robust to noise and nonlinear relationships. A key advantage is its built-in measure of feature importance, which quantifies how much each variable reduces impurity across all trees. This provides insight into which physicochemical properties most strongly influence wine quality, making Random Forest both powerful and interpretable for this task.A key advantage of Random Forest is that it provides a built-in measure of feature importance. After training, the model quantifies how much each variable contributes to reducing prediction error across all trees. This gives useful insight into which physicochemical properties have the strongest influence on wine quality, adding interpretability to the modelling process. Overall, Random Forest was an appropriate choice for this task because it handles nonlinear relationships, reduces overfitting through bagging, stabilizes predictions through averaging, and offers interpretable feature importance metrics. These strengths make it a powerful and reliable model for predicting wine quality.Henceforth,Random forest is one of the best-performing learning algorithms. For social scientists, such developments in algorithms are useful only to the extent that they can access an implementation of the algorithm as supported by [9]

3 Results

3.1 Binary Logistic Regression Model

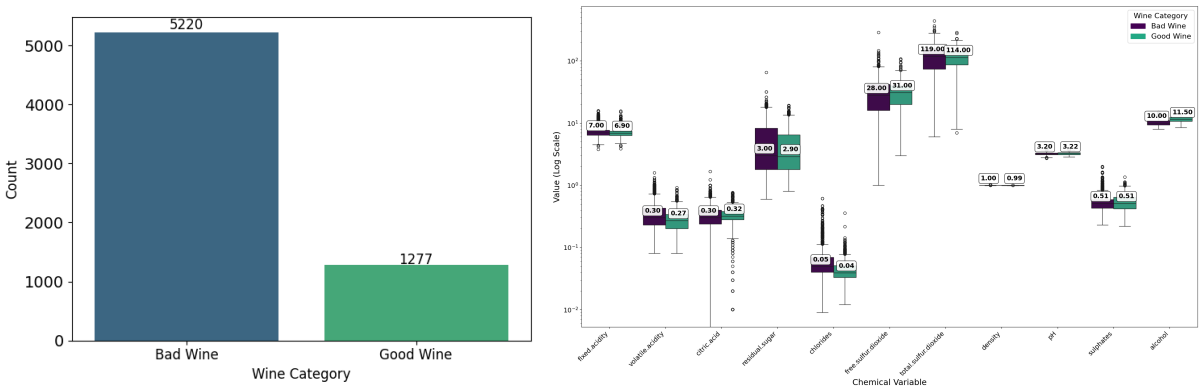
3.1.1 Model Fitting and Summary:

Table 2 highlights the summary of the logistic regression model. The table is important as it shows the significant features that influence the outcome at a significant level.The logistic model was trained to classify wine as Good (Quality ≥ 7) and Bad (Quality ≤ 6).In the preprocessing part, the variables density and free sulfur dioxide were removed to address multicollinearity. The final model was established with the remaining physiochemical properties and the 'type' category. After six successful fisher scoring iterations, several predictors had statistically significant coefficients. Most predictors supported the likelihood of wine being classified as good relatively to being classified as bad. Fixed acidity, volatile acidity, residual sugar, chlorides, pH, sulphates and alcohol possessed significant parameter estimates. The positive coefficient values increases the log odds of wine categorized as good, while the later reduces the log odds.

Table 2: Logistic Regression Summary for Wine Quality Classification

Variable	Estimate	Std. Error	z-value	Pr(>|z|)
(Intercept)	-14.626	1.142	-12.803	< 0.001 ***
fixed.acidity	0.140	0.041	3.397	0.001 ***
volatile.acidity	-4.258	0.382	-11.143	<0.001 ***
citric.acid	-0.479	0.344	-1.393	0.164
residual.sugar	0.047	0.008	5.864	<0.001 ***
chlorides	-11.657	2.676	-4.356	<0.001 ***
total.sulfur.dioxide	0.018	0.010	1.689	0.091
pH	1.006	0.267	3.762	<0.001 ***
sulphates	0.874	0.271	3.618	<0.001 ***
alcohol	0.912	0.038	23.914	<0.001 ***
typeWhite	0.064	0.199	0.323	0.747

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1



(a) Class distribution of Good vs Bad Wine

(b) Distribution of chemical features for Good vs Bad Wine

Figure 2: Comparison of class distribution and chemical feature distributions for Good vs Bad Wine.

3.1.2 Odds Ratios and Confidence Intervals:

The table 3 shows the different variables with their magnitude in influencing the classification outcomes of the logistic regression model.To further investigate the true direction and magnitude of the predictors in the logistic regression, the odd ratios and a confidence interval of 95% were computed for each variable. The odd ratios aid in quantifying how a one unit increase in a predictor might influence the odds of classifying wine to either be Good or Bad. The confidence interval provides us with the degree to which we can trust the influence of the predictors. A good number of chemical physical properties exhibited statistically meaningful relationships. These variables include alcohol and sulphates. These variables increase in one unit, increase the likelihood of classifying wine as

a Good wine. In contrast, volatile acidity and chlorides possess a strong negative relationship with wine quality. This is since their odds ratios fell below 1, indicating that one unit increase in these physiochemical properties greatly reduces the odds of wine to be classified as Good wine. Citric acid,total sulfur dioxide,pH, residual sugar and wine type showed minimal influence. The confidence of these variables included 1, indicating that these physiochemical properties do not have a significant statistical impact on the classification outcomes.

Table 3: Odds Ratios and 95% Confidence Intervals for the Logistic Regression Model

Variable	**Odds Ratio**	**CI Lower**	**CI Upper**
(Intercept)	< 0.001	< 0.001	< 0.001
fixed.acidity	1.15	1.06	1.247
volatile.acidity	0.01	0.01	0.030
citric.acid	0.61	0.31	1.211
residual.sugar	1.07	1.05	1.086
chlorides	< 0.001	< 0.001	< 0.001
total.sulfur.dioxide	0.10	0.10	1.00
pH	2.73	1.62	4.62
sulphates	6.51	3.82	11.06
alcohol	2.49	2.31	2.69
TypeWhite	1.07	0.72	1.58

3.1.3 Classification Report and Confusion Matrix:

Table 4 and table 5 show the summarized report on how best the binary logistic regression model performed on classifying bad wine and good wine from training on both balanced and unbalanced training data.

Table 4: Classification Report and Confusion Matrix for Binary logistic Regression Classifier

Wine Category	**Precision**	**Recall**	**F1-Score**	**Support**
Bad Wine	0.84	0.96	0.89	854
Good Wine	0.58	0.24	0.34	210
Accuracy	–	–	**0.82**	**1064**
Macro Avg	0.71	0.60	0.62	1064
Weighted Avg	0.79	0.82	0.78	1064

Predicted Class		
Actual Class	**Bad Wine**	**Good Wine**
Bad Wine	817	37
Good Wine	159	51

Table 5: Classification Report and Confusion Matrix for Binary logistic Regression Classifier (Balanced with SMOTE)

Wine Category	**Precision**	**Recall**	**F1-Score**	**Support**
Bad Wine	0.92	0.73	0.81	854
Good Wine	0.40	0.73	0.52	210
Accuracy	–	–	**0.73**	**1064**
Macro Avg	0.66	0.73	0.67	1064
Weighted Avg	0.81	0.73	0.76	1064

Predicted Class		
Actual Class	**Bad Wine**	**Good Wine**
Bad Wine	625	229
Good Wine	57	153

3.1.4 Receiver Operating Characteristic Area Under the Curve (ROC AUC) for unbalanced data

The figure shows the comparison of the logistic regression with a random classifier. The random classifier is labeled by the dot lines and the colored lines represent classifications done by the logistic regression model.

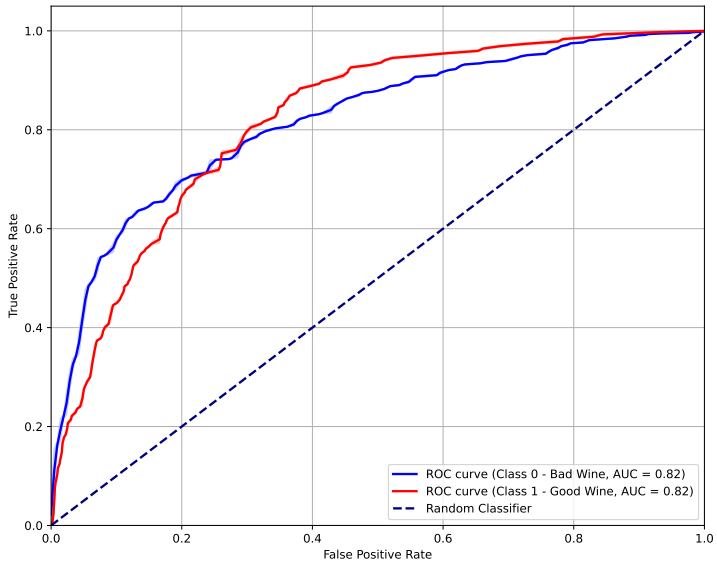


Figure 3: Receiver Operating Characteristic (ROC) Curve for Wine Quality Prediction (Per Class)

3.1.5 Observations:

1. logistic regression model under Unbalanced training data

   The logistic regression summary table indicates variables that are not significant in predicting the classification class. These variables include citric acid, type white and total sulfur dioxide.This true fact when considering the results of the odds ratio.The confidence interval of these variables mentioned above is also not statistically significant .The classification report shows the model better performance in predicting Bad wine than Good wine. Bad wine has a precision of 0.84 and a recall of 0.96, this indicates that the model makes most correct prediction in terms of classifying Bad wine and rarely making few false positives.On the other side, Good wine has a precision score of 0.58 and a recall of 0.24. This is a clear indication that the models performs poorly in predicting Good wine and in a lot of circumstances it classifies Good wine as Bad wine.

   The ROC AUC curve compares the performance of the logistic regression with a dummy classifier. It can be observed that areas under the curve is the same for both classes and a consistency of the model's better performance of classifying Bad wine. The logistic model clearly analysizes the chemical properties that distinguish Good wine and Bad wine.The model has cemented the physiochemical properties that affect highly affect sensory perception, specifically the strong combinations of alcohol, sulphates and chlorides. The other chemical properties have less or no significant effect to the classification of wine. Regardless of the high or low impact towards classification, it is important to balance the all chemical properties in order to achieve the best wine.

2. Binary logistic regression under balanced training data

   The model's performance increased on the recall of good wine from 0.24 to 0.73, but the bad wine recall dropped from 0.96 to 0.73. There is no doubt that the model is better at identifying bad wine, but the tradeoff is inevitable. The confusion matrix cements thus further, it shows the model has improved on classifying good wine but has started slacking off in identifying bad wine. The macro F1-score has also increased from 0.62 to 0.67, an evidence of the model increased capacity on identifying minority class.

3.1.6 Limitations of the model:

The model has managed to identify important patterns in the data and variables that significantly influence wine classification.The model has proven its limitations and these need to be acknowledged. First, The effect of having a misbalance in the dataset has led the model to consistently favour Bad wine over Good wine.This has been proven by the classification report, confusion matrix and the ROC Curve of the unbalanced side. Secondly, the model does not assume non-linear relationships among the variables. The variables in our dataset such as alcohol and sulphates might poses a non-linear relationship which can help the model in training and eventually better classification performance. This is the major reason that has pushed further analysis of considering decision based models as non-linearity relationships among variables is learned by these models.

3.2 Decision Tree Classification

3.2.1 Model Description, Baseline and Tuning

Given the limitation of the binary logistic regression in exploring non-linear relationships and interactions among the varibles, the best next alternative is the decision tree classifier. The decision tree does not consider the assumption of linearity or additive factor among the variables. The decision tree splits the data into groups that share similar characteristics and utilizes the method of recursive splitting.This approach enables the model to learn complex patterns, non-learning relationships and interactions among the physiochemical properties of the data.

A baseline decision tree model was fitted to the data with its predefined defaults. The baseline model performed poorly and exhibited signs of overfitting, the baseline model had training-root mean squared error(rmse) of 0 and a test-root mean squared error (rmse) of 0.22. To address this, tuning was performed using randomized search algorithm. The randomized search was applied on key hyperparameters over a predefined values of each. These hyperparameters included `max_depth`, `min_samples_split`, `min_samples_leaf`, `max_features`, and `ccp_alpha`. Different combinations of hyperparameters were sampled and the final model considered an random selection of hyperparameters that produced a model that performs better on unseen data. The randomized search best result were `max_depth`:5, `max_features` :None, `min_samples_leaf` :19 and `min_samples_split`:24

3.2.2 Feature Importance

The decision tree algorithm provides a natural way of identifying physiochemical properties that directly influence wine classification. The variables that have greater feature importance are based on the fact that their splits lead to greater reduction in impurity. The top ranked important features ranked by the decision tree are always close to the root node and given our case, these variables were alcohol and chloride.The Figure 4b below shows the top 10 features which are utilized to reduce impurity in splitting decisions.The predictors are accompanied by a 'num' prefix due to the application of the pipeline that handles both categorical and numerical variables.

3.2.3 Classification report and Confusion Matrix

The report on the performance of the decision tree classifier is summarized in Table 6. The classification report is left-hand side (a) and the confusion matrix is on the right-hand side (b)

Table 6: Classification Report and Confusion Matrix for Decision Tree Classifier

Wine Category	**Precision**	**Recall**	**F1-Score**	**Support**
Bad Wine (0)	0.83	0.97	0.89	854
Good Wine (1)	0.59	0.18	0.28	210
Accuracy	–	–	**0.81**	**1064**
Macro Avg	0.70	0.60	0.62	1064
Weighted Avg	0.78	0.81	0.77	1064

	Predicted	
Actual	**Bad Wine (0)**	**Good Wine (1)**
Bad Wine (0)	828	26
Good Wine (1)	172	38

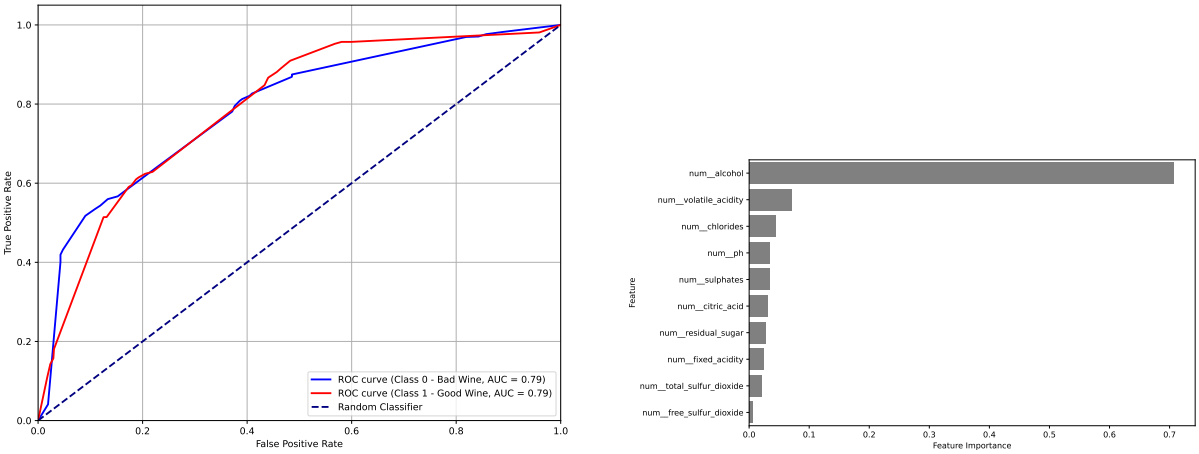
Table 7: Classification Report and Confusion Matrix for Decision Tree Classifier (Balanced with SMOTE)

Wine Category	Precision	Recall	F1-Score	Support
Bad Wine (0)	0.87	0.83	0.85	854
Good Wine (1)	0.43	0.50	0.46	210
Accuracy	–	–	**0.76**	**1064**
Macro Avg	0.65	0.67	0.66	1064
Weighted Avg	0.78	0.76	0.77	1064

	Predicted Class	
Actual Class	**Bad Wine (0)**	**Good Wine (1)**
Bad Wine (0)	713	141
Good Wine (1)	104	106

3.2.4 Receiver Operating Curve (ROC)

The ROC curve shows that the decision tree classifier performs better than a random classifier. The Area under the curve (AUC) appears to be the 0.79 for both Bad wine and Good wine. The ROC curve is on the left-hand side (a). The bar graph of feature importance is on the right-hand side (b).



(a) ROC AUC Curve for the Random forest Classifier

(b) The top 10 feature importance for the Decision Tree

Figure 4: The ROC curve and Feature Importance bar graph of the Decision Tree Classifier

3.2.5 Observations

1. Decision tree classifier model under Unbalanced training data

   The Decision tree classifier has performed a little less than the binary logistic regression by achieving an accuracy of 0.81.The model has achieved 0.83 precision and 0.97 recall on Bad wine, this shows the promising ability of the model in identifying Bad wine and few cases of misclassification. On the later side, a recall of 0.18 shows that the model is struggling a lot in classfying the minority class. The Decision tree classifier has been able to identify only 59% of Good wine. The F1-score of 0.28 is terrible as it indicates that the model is not really good in predicting high quality wine. The ROC curve shows that the decision tree classier is still better over a model that predicts the outcome randomly regardless of the class imbalance within the wine quality dataset.The area under the curve is 0.79, it is a good value but does not really show the ability of the model in predicting both classes individually.

2. Decision tree classifier model under balanced training data

   The model increased its capability in identify good wine, from a rise of the recall from 0.18 to 0.76. Table 7 shows an improvement of classifying good wine from 38 to 106. A trade-off can also be observed from the drop of bad wine recall and F1-score. The macro avg F1-score has slightl increased from 0.62 to 0.66, indicating a more balanced emphasis on both classes than the later.

3.2.6 Limitations

Decision tree classifier is a white box, this is due to the fact that it is easy to interpret the results. However, the decision trees are prone to overfitting, learning so much on the training data and fail to generalize on unseen data.Additional, decision trees embody variance, a change in the training data can lead to changes in tree structure and predictions. To address such concerns, Random forest has been introduced in this study to tackle variance and noise from the training data.

3.3 Random forest Classifier

3.3.1 Model Description, Baseline and Tuning

The Random forest classifier is an extension of decision tree and it is categorized as ensemble method. Ensemble methods are highly regarded in the data mining field as they combine different algorithms to make predictions. The Random forest is built on top of multiple trees and it utilizes bootstrapping technique in sampling data. The trees are trained on data which is of the same size as the original dataset. The technique is successful due to replacement that is done on the original dataset that enables all the trees to get unique training dataset. The trees are tested on data that has not been seen and this excludes validation sets. Bootstrapping introduces variety as each tree obtains different data and noise from the data is greatly reduced. The model utimately performs bagging which combines predictions of all trees and in that effort the variance is reduced due to the output considerations of all trees.

A baseline model was fitted on the training data with the default parameters. Out of the bag(OOB) tuning criterion was utilized to estimate the optimal number of trees $(n_estimators)$ with minimum OOB error.Different Random forest models were trained with a range of 10 to 500 trees with the goal of assessing the OOB error. The Figure 4b shows a declining trend of the OOB error as the trees increase. The team select 260 as the number of trees, since going forth the variance was less significant. The final Random classifier model used 260 trees following that significant fact.

3.3.2 Feature importance

The Random forest algorithm utilizes a robust estimation of feature importance in comparision to a single tree. Feature importance estimation is based on the total mean decrease in impurity for the given variable. The algorithm calculates the total reduction in Gini impurity through all splits done by a variable in all trees and given the result the ranking proceeds. The Figure 5 shows that alcohol, density and chrolides are the top 3 which is slightly different from the decision tree classifier.The least important physiochemical properties are $type_white$ and $type_red$, which are the categories of wines in our dataset. The ditribution of feature importance is much better than decision tree classifier, as features are given more balanced feature importance.

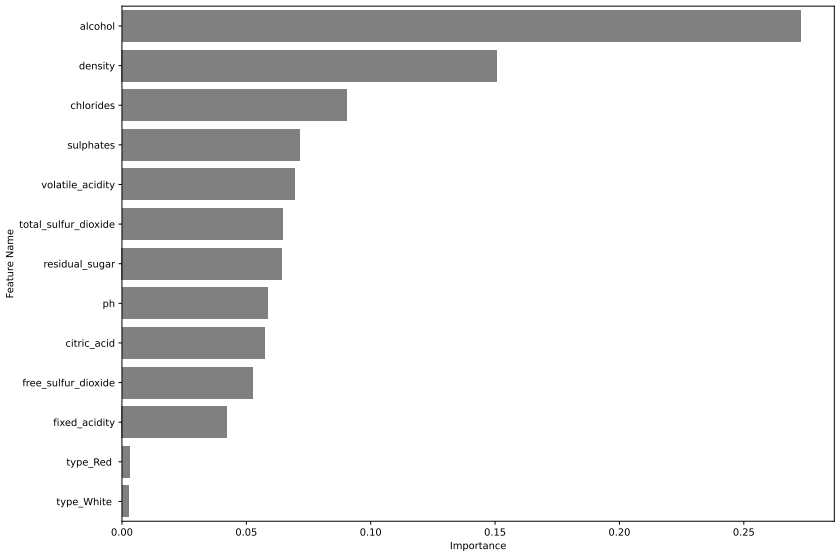


Figure 5: The top 10 feature importance of Random forest classifier

3.3.3 Classification Report and Confusion Matrix

The Random forest classifier's performance is summarized by two tables below for both classes of wine. The classification report on the left-hand side and the confusion matrix on the right-hand side

Table 8: Classification Report and Confusion Matrix for Random Forest Classifier

Wine Category	**Precision**	**Recall**	**F1-Score**	**Support**
Bad Wine (0)	0.84	0.97	0.90	854
Good Wine (1)	0.65	0.24	0.35	210
Accuracy	**–**	**–**	**0.82**	**1064**
Macro Avg	0.74	0.61	0.63	1064
Weighted Avg	0.80	0.82	0.79	1064

Predicted Class		
Actual Class	**Negative (0)**	**Positive (1)**
Negative (0)	825	29
Positive (1)	159	51

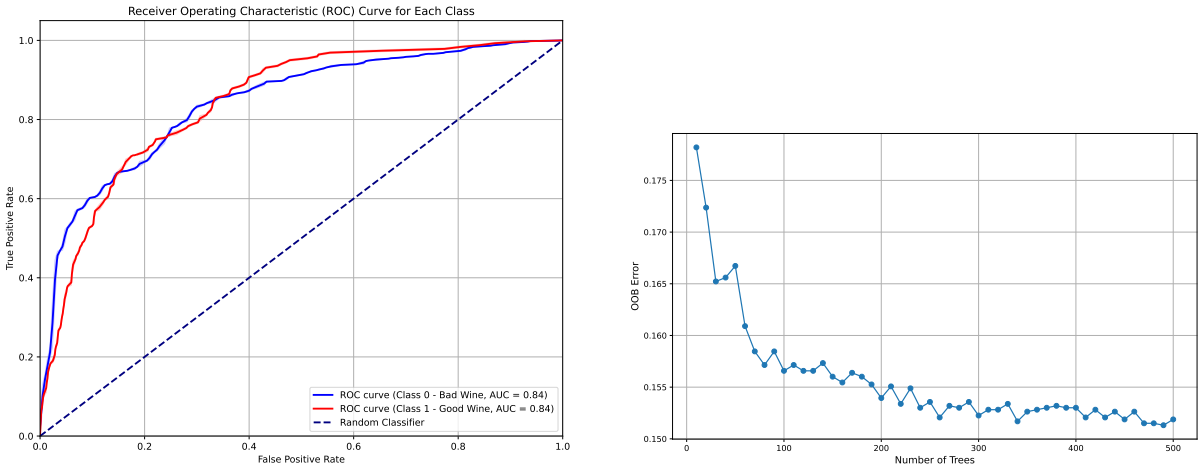
Table 9: Classification Report and Confusion Matrix for Random Forest Classifier (Balanced with SMOTE)

Wine Category	**Precision**	**Recall**	**F1-Score**	**Support**
Bad Wine (0)	0.91	0.80	0.85	854
Good Wine (1)	0.46	0.67	0.54	210
Accuracy	**–**	**–**	**0.78**	**1064**
Macro Avg	0.68	0.74	0.70	1064
Weighted Avg	0.82	0.78	0.79	1064

Predicted Class		
Actual Class	**Bad Wine (0)**	**Good Wine (1)**
Bad Wine (0)	687	167
Good Wine (1)	69	141

3.3.4 Receiver Operating Curve (ROC) AUC and Out of the Bag Curve (OOB) for unbalanced data

The comparison of the Random forest classifier against a random classifier can be seen at Figure 6a on the left-hand side.The Figure 6b on the right-hand side shows OOB curve exhibiting a downward trend.



(a) ROC AUC Curve for the Random forest Classifier

(b) OOB for the Random Forest Classifier

Figure 6: The ROC AUC and OOB Curve of the Random classifier

3.3.5 Observations

1. Random forest model under unbalanced training data

   The Random forest classifier has performed better than decision tree classifier achieving an overall accuracy of 0.82. The model is highly sensitive to classification of Bad wine, suggested from having a recall of 0.97. The Random forest classifier is also performing better at classifying Bad wine from Good wine, as has a preciion of 0.84. In classifying Good wine, the model has improved relatively from the decision tree classifier. The precision score of 0.65 and recall of score 0.24 rivals the non-ensemble method in terms of performance.This is a clear indication of the advantages of combination of different trees which leads to less variance and reduction in noise in the training data. The ROC curve proves that the Random forest classifier is a good model relatively to a model that predicts classes randomly. The areas under the curve (AUC) appears to be 0.84 for both classes.

2. Random forest model under balanced training data

   The final model trained on balanced training data maintained a good balanced in identifying bad wine and good wine. The good wine recall increased by a magnitude of 32 units, while the later recall drop by 14 units. The trade-off is still persistent in all models, but the random forest classifier managed to better identify both class than all other models. The Random forest was better at identifying good wine and slightly missed a few bad wine, as show by table 9

3.3.6 Limitations

Despite the better performance of the Random classifier in prediction, the model has wrongly identified 159 as Bad wine. This is due to the fact of class imbalance within the dataset, there are more observations of low quality wine than good quality wine. The Random classifier model is learning a lot from Bad wine (majority class) and it is struggling to identify Good wine (minority class). Additionally, the model has a good distribution of feature importance relatively to the decision tree classifier but it appears still be dominated by small subset of predictors. These predictors are alcohol and density, such a case limitations the full exploitation of other predictors which have less feature importance.

4 Discussions

The results from the different models indicate a significant difference in predictive power, interpretability and effectiness.The Binary logistic model did not perform well, this is due to its linear relationship assumption. It underperfomed because of not considering non-linear relationships among the physiochemical properties of wine. However, it was the simplest and easiest model to interpret. The decision tree classifier improved classification performance and the hierarchical splits were heavily based on alcohol and acidity. Due to baised splits, the Decision tree classifier was effective in identifying bad wine and worse on identifying good wine. The baised splits were a result of sensitivity to noise and overfitting. The Random forest was the best model overall, specifically in the balanced training data category. The Random forest was introduced to tackle variance and unbalanced feature importances ranking of the physiochemical variable. The Random forest produced a more balanced in predictive capability for both the class, possessing a macro avg F1-score of 0.70. The findings of this study are consistent with the previous research of wine predictive modelling[4], cementing the fact that ensemble tree methods perform better than single classifiers in classifications task of predicting wine quality.

All the models struggled in identifying good wine, especially Binary logistic regression and Decision tree classifier. The dataset imbalance acting as the main factor of their ineffectiveness. The application of smote increased performance of the models in identifying good wine by improving the recall of good wine but inevitably introduced a trade-off of precision for both classes. Box plot 2b revealed that good wine is associated with higher levels of alcohol, and lower levels of both volatile acidity and chlorides. This is a significant fact, since it aligns perfectly with established oenological literature.

5 Conclusions and Future scope

This study demonstrates that it is possible to utilize machine learning techniques in modelling physiochemical attributes of wine in predicting the sensory quality scores. Binary logistic regression, Decision tree and Random forest models exhibited varying degrees of accuracy and generalization performance, but the Random forest was the best model overall. The superiority of the Random forest model was based on the best macro avg F1-score of 0.70 and its shield against noise. Its characteristics of introducing randomness and including a lot trees makes it the best approach among all the three models in this study. The results confirm that models with the capability of learning non-linear relationships and entail ensemble approach are best suited for understanding complex chemical interaction relationships. Additionally, the study reveals that there are meaningful patterns in the data, but wine classification task remains difficult due to the aspect of subjectivity of scoring wine quality.

The future tasks entail exploring more advanced ensemble algorithms that have performed better than traditional machine learning models such as Random forest. The advanced algorithms include Gradient boosting, XGBoosting, LightGBM or Artificial Neural network (ANN). it is interesting how the existing research has extended the study to further explore predicting geographical locations of wine categories[10], This is another proof of application of advanced algorithms in this field. Machine learning provides varying options of enhancing predictive stability and performance, these include probabilistic calibration, cross model stacking and hyperparameter tuning which will be highly considered.

The class imbalance remains a critical area for exploration.The scientific study of *Analysis of Multiclass Imbalance Handling in Red Wine Quality Dataset Using Oversampling and Machine Learning Techniques*[11] and *Wine feature importance and quality prediction: A comparative study of machine learning algorithms with unbalanced data*[12] provide various alternatives of utilizing smote in handling class imbalance. The future scope will consider the best approach of handling class imbalance from these two studies. The study will also considering adding more samples of high quality wine which is a better alternative to synthetic data, this will definitely improve model generalization.

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